

8 (**short**) A small ball is rolling without friction with initial velocity v on a track which is initially horizontal and then goes around in a vertical loop of radius R , as shown in Fig. 5. You can assume that the rotational kinetic energy of the ball is negligible.

(a) Calculate the speed of the ball as a function of the angle θ as it goes around the loop. [5]

(b) What is the minimum initial speed required for the ball not to lose contact with the track ? [5]

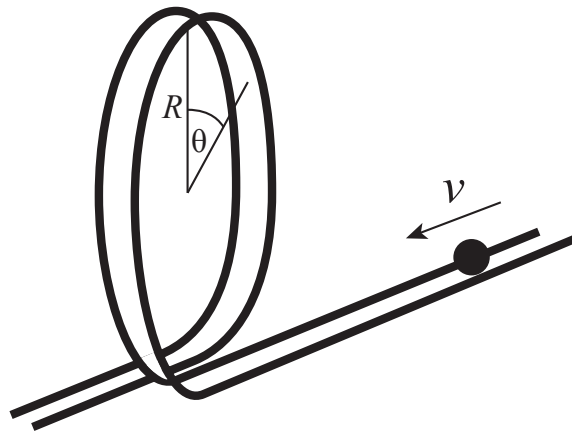


Fig. 5